

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1211

Pokagon 100 kV
May 31, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1211. The Interconnection Customer proposes a 170.3 MW solar photovoltaic facility interconnecting at the Pokagon 100 kV substation in the Indiana area. The request was received on May 22, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 468 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,410,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: DENVER TERMINAL / POKAGON 100kV
Contingency: BONNET CREEK / POKAGON 100kV
Loading: 126.0% (Rating: 126 MVA)
DFAX: 100.0%
Pre-project loading: 9.7%
- Facility: BONNET CREEK / POKAGON 100kV
Contingency: DENVER TERMINAL / POKAGON 100kV
Loading: 120.3% (Rating: 132 MVA)
DFAX: 100.0%
Pre-project loading: 9.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DENVER TERMINAL / POKAGON 100kV	Rebuild line with higher-capacity conductor	\$5,128,000	100% (\$5,128,000)
BONNET CRK / POKAGON 100kV	Rebuild line with higher-capacity conductor	\$5,282,000	100% (\$5,282,000)
	Total allocated to project		\$10,410,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.